

DATABASE MANAGEMENT SYSTEMS (DBMS) COURSE PROJECT

IT Helpdesk & Asset Support Management System

Presentation-I: Problem Description, Domain Context, Objectives & Scope Specification

PRESENTED BY

Md Aali Rahman

Roll Number: **25WU0102156**

Section: AIML Panthers | Academic Year: 2026

Woxsen University, School of Technology

ACADEMIC GUIDANCE

Dr. Kiranmayee Adavala

Faculty Guide | School of Technology

Course: Database Management Systems

Relational Engine: MySQL 8.0+ / 9.x (InnoDB)

IT Service Operations & Existing Operational Bottlenecks

ENTERPRISE CONTEXT

Modern IT Infrastructure

- Educational institutions and enterprises deploy hundreds of diverse computing assets (workstations, laptops, network switches, peripherals).
- Continuous IT support requests are raised daily by faculty, administrative personnel, researchers, and students.
- Operations require coordination between multi-tiered support staff, hardware inventory, and manufacturer warranties.
- Seamless service delivery directly underpins institutional productivity and technical teaching workflows.

CURRENT DEFICIENCIES

Fragmented Management

- Widespread reliance on unstructured emails, spreadsheets, and informal communication channels.
- Tickets are frequently lost, delayed, or untracked across busy technical shifts.
- Lack of clear technician assignment leads to duplicate efforts or unattended high-priority outages.
- Hardware movements occur without updating user ownership or tracking warranty expiries.

Formal Problem Statement & Academic Motivation

PROBLEM FORMULATION

The Relational Challenge

- The absence of a centralized, normalized relational database for IT service operations causes severe data redundancy, inconsistent state lifecycles, and untracked repair costs.
- Manual spreadsheets lack ACID compliance, fail to enforce foreign key referential integrity, and provide zero immutable audit trails for ticket status changes.
- There is no structured differentiation between unplanned incident disruptions and routine service requests.

ADVERSE CONSEQUENCES

Institutional Impact

- Prolonged Mean Time to Resolution (MTTR) due to lost ticket historical context.
- Financial leakage from paying out-of-pocket for equipment repairs that are covered under valid manufacturer warranties.
- Zero visibility into department-wise IT resource consumption and technician workloads.
- Compromised audit compliance and operational accountability across technical shifts.

Centralized MySQL IT Helpdesk & Asset Support System

RELATIONAL CENTRALIZATION

Unified Relational Model

- Engineered with a 14-table Third Normal Form (3NF) relational database in MySQL 8.0+ / 9.x.
- Enforces strict referential integrity with ON DELETE RESTRICT, CASCADE, and SET NULL constraints.
- Eliminates data redundancy across departments, users, assets, categories, and technicians.
- Guarantees ACID transactions for all status updates and asset reassignments.

ARCHITECTURAL PILLARS

Key Functional Subsystems

- **Decoupled Ticketing:** Clear subtype classification into Incidents (outages) vs Service Requests (provisioning).
- **Hardware Lifecycle & Warranty:** Asset tagging, serial number tracking, and repair expenditure accounting.
- **Immutable Status Audit Trail:** Automatic timestamped tracking of all ticket status progressions.
- **Multi-tier Access:** Both terminal CLI and responsive modern web interface connected to MySQL.

Project SMART Objectives

TICKETING & QUEUES

Centralize Support

- Standardize ticket logging across 5 severity levels and 5 operational categories.
- Provide clear ownership and tracking from initial filing to verified resolution.

LIFECYCLE AUDITING

Immutable Audit Logs

- Maintain a full chronological status history for every ticket (Open → In Progress → Resolved → Closed).
- Record transition timestamps and previous states for accountability.

ASSETS & INTEGRITY

Hardware Governance

- Link physical assets with users, warranty validity, and repair expenditures.
- Enforce strict relational constraints (PK, FK, UNIQUE, CHECK) across all 14 tables.

System Scope & Operational Boundaries

IN-SCOPE MODULES

Core System Scope

- Master department and user records with verified institutional emails.
- Ticket lifecycle management: categorization, prioritization, assignment.
- Specialized subtype tracking for Incidents vs Service Requests.
- Asset inventory with unique asset tags, serial numbers, and warranty contracts.
- Maintenance expense tracking and total cost calculation per department.
- Relational reporting: JOIN queries, aggregate workloads, audit timelines.

EXPLICIT EXCLUSIONS

Operational Boundaries

- Commercial payment gateway processing (maintenance costs are accounting logs).
- Automated machine learning ticket auto-triage (reserved for future work).
- Direct hardware telemetric scanning or IoT board firmware flashing.
- Public open internet self-registration (user directory is managed by IT admin).

System Users, Roles & Responsibilities

ROLE 1

End-Users

- Employees, faculty members, and students.
- Raise new support tickets with title, description, category, and priority.
- Classify request as an Incident or Service Request.
- Track real-time resolution progress and view allocated hardware assets.

ROLE 2

Support Technicians

- IT engineers categorized by technical specialization.
- Accept assigned tickets and update status to In Progress.
- Investigate faults and record official resolution narratives.
- Inspect asset maintenance history and warranty coverage.

ROLE 3

Helpdesk Managers

- IT administrators and systems leads.
- Monitor queue metrics, active tickets, and technician workloads.
- Register new hardware assets, warranty contracts, and vendors.
- Log maintenance repair invoices and generate cost reports.

Major Functional Requirements (FR1 – FR7)

TICKETING FLOW

FR1 – FR3

- **FR1 (Ticket Creation):** Users create tickets with unique ticket_no, category, priority, and subtype.
- **FR2 (Assignment):** Administrators assign pending tickets to specialized support staff.
- **FR3 (Status Lifecycle):** System logs every state change into status_histories with old/new states.

RESOLUTION & ASSETS

FR4 – FR5

- **FR4 (Resolution):** Technicians record resolution descriptions and timestamps with unique 1:1 enforcement.
- **FR5 (Asset Tracking):** Hardware assets cataloged with unique asset_tag, serial_no, custodian user, and warranty.

MAINTENANCE & SQL

FR6 – FR7

- **FR6 (Maintenance Logging):** Record asset repair dates, descriptions, and expenses with check constraints.
- **FR7 (Relational Reporting):** Provide multi-table JOINS, workload summaries, and audit trail extractions.

Expected Operational Benefits & Impact

OPERATIONAL EFFICIENCY

Streamlined Workflows

- **Faster Turnaround:** Eliminates lost tickets and clarifies technician ownership through automated queues.
- **Automated Accountability:** Complete audit trails deter bottlenecks and track resolution performance.
- **Reduced Redundant Inquiries:** Structured search and categorization prevent duplicate ticket filings.
- **Data Integrity:** Foreign key cascades eliminate orphan records when tickets or assets are modified.

RESOURCE OPTIMIZATION

Hardware & Financial Governance

- **Zero Lost Equipment:** Accurate asset-to-user mappings prevent misplaced or unreturned laptops.
- **Warranty Protection:** Immediate visibility into vendor warranty dates prevents paying for covered repairs.
- **Budget Forecasting:** Departmental maintenance reports highlight recurring equipment failures.
- **Informed Procurement:** Historical failure logs guide future enterprise hardware purchases.

Conclusion & Academic Road Map

REVIEW 1 ACCOMPLISHMENTS

Requirements Baseline

- Comprehensive problem analysis and stakeholder identification completed.
- Clear boundaries and functional requirements (FR1 – FR7) established.
- Conceptual relational entities identified for 14 tables in 3rd Normal Form.
- Full alignment with DBMS course learning outcomes and project specifications.

SUBSEQUENT EVALUATION PHASES

Presentation–II & III Road Map

- **Presentation–II Deliverables:** Complete MySQL DDL schema, 55 relational seed records, ER diagram, and analytical queries.
- **Presentation–III Deliverables:** Live working user interface connected to MySQL demonstrating real-time INSERT, DELETE, and VIEW operations.
- **Final Project Report:** Comprehensive 16–chapter academic documentation.